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ARTICLE

Efficient CO_2 Capture from Ambient Air with Amine-Functionalized Mg–Al Mixed Metal Oxides

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The search of efficient amine-functionalized materials for direct air capture (DAC) of CO_2 is highly attractive to achieve negative emissions. Here we report a new family of class 1 adsorbents for DAC by loading 67 wt.% branched poly(ethylenimine) (PEI) into Mg-Al-CO_3 layered double hydroxide-derived mixed metal oxides (MMOs), which exhibit unexpectedly large CO_2 uptakes (2.27 mmol g^{-1}), fast kinetics (1.1 mmol g^{-1} h^{-1}), and high stability over 20 cycles at 25 °C under 0.4 mbar CO_2. The prepared MMO nanosheets are self-assembled into spherically shaped particles with abundant slit-like mesopores and broad pore distribution. This ideal nanostructure leads to the uniform distribution of the impregnated PEIs and optimal accessibility for CO_2 molecules. The defect-abundant MMO layers provide strong electrostatic attraction to the attached polyamines, making the hybrid adsorbents thermally robust up to 300 °C. Enhanced capacity and stability are observed in the presence of moisture, thus facilitating the use of steam stripping to concentrate the desorbed CO_2.

Introduction

Direct air capture (DAC) is a type of negative emissions technology that extracts CO_2 from the atmosphere via absorption or adsorption. Compared to conventional CO_2 capture technologies that are used to remove CO_2 from large point sources, DAC is advantageous because it could be used for distributed CO_2 emissions, it presents no location restrictions for the placement of the capture facilities, and its performance is not affected by contaminants, such as SO_x, NO_x, and heavy metals.[1] Furthermore, when DAC is coupled with CO_2 chemical conversion technologies it could provide carbon sources for the production of fuels, biofuels, pharmaceuticals, high-value chemicals, greenhouse feeding nutrients, and agricultural.[2–6] Early DAC systems have used aqueous alkali or alkali-earth hydroxides, which extracted CO_2 via causticization or alternative causticization.[7,8] However, the high energy demands for their thermal regeneration, large amounts of water loss, and use of pure O_2 have limited their implementation, and therefore, researchers changed their focus to adsorbent-based systems.[9,10] DAC technologies have currently reached technology readiness level 7 (demonstration level).[11] Carbon Engineering estimated that the cost of DAC demonstration plants using an aqueous KOH sorbent is 94–232 \$/t.[12] A recent work indicates that for adsorbent-based systems, the minimum energy requirement of DAC is 0.113–0.145 MJ/mol-CO_2 and the cost ranges from 60 to 190 \$/t.[13] Further cutting down the cost of DAC can be achieved through the development of adsorbent materials.[14,15]

When selecting appropriate adsorbents for DAC one should comprehensively consider their cost, adsorption capacity, kinetics, multi-cycle stability, and selectivity. In addition, the binding force between CO_2 and adsorption sites should be strong enough because the CO_2 in the atmosphere is extremely dilute (approximately 400 ppm, which is roughly 1/350 of the CO_2 concentration in flue gas). Therefore, solid-supported amines are suitable DAC adsorbents owing to their chemisorption properties and low regeneration temperatures (commonly below 120 °C).[16,17] Supported amines react with CO_2 to form ammonium carbamate,[18] ammonium carbonate,[19] and bicarbonate.[20] Amines could be physically loaded (class 1), covalently tethered (class 2), or *in situ* polymerized (class 3) onto solid supports.[21] Class 1 adsorbents, which is prepared by impregnating amines into mesostructured supports have become highly attractive for commercial-scale DAC applications because of their low cost and relatively large CO_2 uptakes under ultradilute conditions.[11,14,22–29] Low molecular weight primary amine-rich branched poly(ethylenimine) (PEI) is commonly used for class 1 adsorbents, as smaller amines lead to considerable evaporation and amine loss,[30] whereas high molecular weight PEI could hinder the diffusion of CO_2 into the porous matrix.[31]

Although it is believed that amine loading mainly determines the CO_2 uptake of solid-supported amine materials, the textural properties and surface microstructure of the support change the morphology of amines within the support and the gas-surface interaction, and consequently, significantly affect their overall CO_2 capture efficiency. Silica materials, such as commercial silica,[32] fumed silica,[31,33] SBA-15,[34–39] silica fiber,[23] pore-expanded MCM-41 (PE-MCM-41),[40] and mesocellular foam (MCF),[41] have been widely studied for DAC applications owing to their large surface

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areas and high pore volumes. Despite exhibiting promising CO_2 uptakes and amine efficiencies, PEI/silica composites present kinetics and thermal stability drawbacks. Choi et al.[32] reported that 45 wt.% PEI-impregnated commercial silica reached the high CO_2 uptake of 2.36 mmol g^{-1} during the first adsorption cycle (400 ppm CO_2/Ar, 25 °C), but a long adsorption half-time of 309 min was needed and 30% of the CO_2 uptake was lost after only four cycles. Another evidence is that for highly amine-loaded PEI/silica, elevated temperatures were needed to achieve the maximum CO_2 uptake because of the kinetic limitation.[31] The chemical (oxidative degradation and CO_2-induced deactivation) and hydrothermal stabilities of PEI/silica composites are also challenging, and this is particularly important when steam stripping is used in DAC systems to produce concentrated CO_2 streams.[42,43]

To address the above-mentioned drawbacks, it is necessary to design support materials with suitable morphology, strong affinity to polyamines, and high thermal and hydrothermal resistance. It has been reported that both the stability and CO_2 adsorption kinetics of PEI/silica composites could be improved via the addition of 3-aminopropyltriethoxysilane and tetraethyl orthotitanate, because these compounds strengthen the bonding between PEI and the surface of silica.[44] The co-addition of polyethylene glycol to PEI/silica composites could also increase their air capture performance owing to the improved arrangement of PEI with scattered aggregates.[38] Support modification is another strategy that could be used to improve the CO_2 adsorption efficiency of PEI/silica composites under ultradilute streams. Kuwahara et al.[34,35] demonstrated that incorporating heteroatoms into SBA-15 changed the acid/base properties of the support, and thus increased not only the CO_2 uptake but also the regenerability and thermal stability of the adsorbent. However, a later study indicated that the improved CO_2 adsorption properties were more likely related to the changes in texture of the supports (larger pore volume, pore size, and less microporosity).[36] Sayari et al.[40] determined that excellent amine efficiency and stability could be achieved using PE-MCM-41 with abundant surface cetyltrimethylammonium ($CTMA^+$) cations. They concluded that the $CTMA^+$ layer effectively changed the dispersion of PEI on the surface of silica and thus increased amine accessibility. Several researchers used non-silica supports, such as mesoporous carbon and γ-alumina. The use of mesoporous carbon frameworks and additives promoted both pore and amine diffusion, which led to remarkable kinetics and stability at high PEI loadings.[45] PEI-loaded alumina also exhibited high stability under steam striping, but its CO_2 uptake was lower than that of PEI/silica composites.[46,47]

In this paper, we report a new family of class 1 adsorbents for DAC; these adsorbents were obtained by loading PEI into layered double hydroxide (LDH)-derived mixed metal oxides (MMOs). LDHs are two-dimensional layered materials comprising brucite-like layers separated by charge-compensating anions and interlayered water, and their general formula is $[M_x^{2+}M^{3+}(OH)_{2x+2}]^+(A^{n-})_{1/n}\cdot mH_2O$. Each metal cation is coordinated with six –OH groups, which are edge-sharing, to form infinite 2D sheets. Stacked bulk LDHs could be exfoliated into small plates using the aqueous miscible organic solvent treatment (AMOST) method, which was recently developed by O'Hare's group.[48,49] Their large specific surface area, ultrathin nanosheets, and tunable metal cations render exfoliated LDHs usable for catalysts, adsorption, biomedicine, water treatment, and electrochemistry.[50] Upon thermal treatment, LDHs undergo dehydration, dehydroxylation, and decarbonation processes and form MMOs with abundant surface defects.[51] The use of MMOs as supports allowed the development of ideal CO_2 diffusion channels through the slit-shaped mesopores formed by the exfoliated nanosheets, and the extra pores derived from the release of gas molecules, such as CO_2 and H_2O, during calcination. Impregnated polyamines are supposed to disperse well on defect-abundant MMO layers owing to their strong electrostatic attraction even in the absence of additives. Furthermore, the morphology and chemical properties of MMOs could be adjusted by changing the type and relative ratios of divalent and trivalent metal cations, which facilitated the optimization of the CO_2 capture efficiency of the PEI/MMO composites.

Experimental

Synthesis of MMOs

We synthesized MMO precursors using co-precipitation combined with AMOST. All chemicals used for the synthesis were of analytical purity. Typically, 1 M metal precursor ($Mg(NO_3)_2\cdot 6H_2O$ and $Al(NO_3)_3\cdot 9H_2O$) solution was slowly added to an equal amount of 0.5 M Na_2CO_3 solution under vigorous stirring, and the pH was adjusted to 10.0 ± 0.1 by adding NaOH solution to the mixture. The mixture was then aged at 25 °C for 16 h followed by adequate flushing with deionized water. The key step for the AMOST method is the use of aqueous miscible organic (AMO) solvents to rinse and re-disperse the wet sample cake before it is dried. This process was determined to be efficient for the removal of the interlayered water. The AMO solvent (acetone in this study) was then volatilized via overnight vacuum drying to facilitate the exfoliation of the sample nanosheets. The MMOs were obtained via the vacuum calcination at 450 °C for 5 h. The prepared MMO precursors and MMOs were denoted as Mg_xAl-CO_3 and Mg_xAl-O, respectively, where x is the Mg/Al molar ratio.

Preparation of adsorbents

The PEI-based adsorbents were synthesized via wet impregnation. Typically, the MMOs were vacuum dried overnight at 100 °C to remove any adsorbed species. A given amount of branched PEI (Alfa Aesar, Mw ≈ 600) was dispersed into 20 mL methanol. After the dispersion reached equilibrium, 0.5 g dried support was added to it and the mixture was stirred at 25 °C under N_2 protection for 3 h. Methanol was subsequently removed via rotatory evaporation followed by vacuum drying the resulting powder at 60 °C for at least 12 h. Prior to analysis, the samples were vacuum-stored in vials under atmosphere temperature. For comparison, commercially available mesoporous materials, such as SBA-15 (which was purchased from Nanjing XFNANO Materials Tech Co., Ltd), γ-Al_2O_3 (which was procured from D-chem), and light MgO (which was acquired from Sinopharm Chemical Reagent Co., Ltd) were also used as supports. The prepared adsorbents were denoted as PEI*y*/support, where *y* represents the weight ratio of PEI in the PEI/support composite.

Material characterization

The crystal structure of the samples was characterized using a MiniFlex600 (Rigaku) powder XRD instrument with Cu Kα radiation at 40 kV and 15 mA in the 2θ range of 5° to 90°. The morphologies and element distributions of the samples were observed using a MAIA3 (TESCAN) scanning electron microscopy (SEM) apparatus equipped with an EDS detector and a Talos L120C G2 (Thermo Scientific) transmission electron microscope. Prior to the SEM experiments, the powdered samples were sprayed with Au for 30 s. The Mg/Al ratios of the MMO precursors were determined using an iCAP-7600 (Thermo-Fisher) ICP-OES system. The textural properties of the samples were measured via N_2 adsorption at 77 K using an ASAP 2460 (Micromeritics) surface area and porosity analyzer, and the surface area/average pore diameters and pore volumes were evaluated using the Brunauer–Emmett–Teller and Barrett–Joyner–Halenda desorption methods, respectively. The MMO precursors were thermally decomposed in the temperature range of 50 to 800 °C using a Q5000IR (TA Instruments) TGA instrument at the heating rate of 10 °C min^{-1}.

CO_2 adsorption measurement

CO_2 adsorption isotherms in the pressure range of 0−1 bar were acquired using volumetric methods utilizing an ASAP2020 (Micromeritics) surface area and porosity analyzer. Data points were collected after the pressure change was lower than 0.01% over 5 s. The temperature of the samples (25−65 °C) was controlled using a circulator bath. Before each test, samples were vacuum degassed at 100 °C for 5 h.

CO_2 adsorption kinetics and TPD experiments were conducted using a STA449F3 (Netzsch) TGA instrument. Buoyancy effects were corrected and the flow rate of 100 mL min^{-1} was used for all experiments. The balance gas was adjusted to the minimum flow rate (10 mL min^{-1}) to avoid the mixing effect. Approximately 10 mg of each sample was pre-activated under high purity N_2 flow at 120 °C for 1 h. For the kinetics test, the temperature was reduced to 25 °C and the purge gas was switched to 400 ppm CO_2 in N_2 for 12 h. For the TPD test, the CO_2-saturated adsorbents were heated from 25 to 450 °C under N_2 flow at the heating rate of 5 °C min^{-1}.

Stability tests were conducted using a home-made fixed bed, and the details of the experimental set-up are shared in the Supporting Information. Typically, 0.8 g samples were loaded into a stainless-steel tube (inner diameter of 6 mm), and were alternately used for adsorption at 25 °C under 400 ppm CO_2 in N_2 for 2 h followed by desorption at 120 °C under pure N_2 for 1 h. Approximately 3% moisture was added to the gases by passing them through a scrubbing bottle. The CO_2 content of the tail gas after drying using a $CaCl_2$-filled tube was determined using an THA100S non-dispersive infrared analyzer (0−600 ppm, ±2% full scale).

CO_2 adsorption mechanism measurement

In situ FTIR tests in the range of 4000 to 400 cm^{-1} were performed using a Bruker Tensor 27 (Bruker Optik) analyzer at the resolution of 4 cm^{-1}; 32 scans were performed for each sample. Powdered samples were pressed into 12 mg cm^{-2} thick wafers, and then were loaded into a home-made stainless-steel cell with KBr windows. Prior to analysis, the samples were electrically heated to 120 °C for 1 h under vacuum to remove any adsorbed species. After cooling to 25 °C, CO_2 was step-by-step added to the sample cell using a quantitative loop, and the FTIR spectra were recorded at 30 min intervals for each pressure point.

Results and discussion

We first prepared two types of Mg_xAl-CO_3 LDH (x = 2, 3) using the AMOST method, and the metal contents of the samples were determined using inductively coupled plasma optical emission spectrometry (ICP-OES) tests (Table S1, Supporting Information). A compound using Mg:Al in the ratio of 0.55 was also prepared ($Mg_{0.55}Al$-CO_3) (Table S1). All samples exhibited exfoliated morphologies with flower-like structures and slit-shaped mesopores (Fig. 1). The size of the sample nanosheets could be adjusted from 200 to 20 nm by changing x owing to the increase in the content of Al^{3+} ions introducing more defects into the brucite-like MgO structure.[52] The typical LDH patterns of the samples were confirmed using X-ray diffraction (XRD) analysis, and the results are presented in Fig. S1a. The XRD pattern of $Mg_{0.55}Al$-CO_3 suggested that it is a mixed phase of LDH and Al-containing impurities (*e.g.* $Al(OH)_3$). As x decreased the (003) and (006) peaks shifted toward larger angles, which indicated that the interlayer spacing was shortened owing to the stronger electrostatic interactions between the positively charged layers. After calcination at 450 °C for 5 h, all samples were transformed into MMOs (denoted as Mg_xAl-O), and only weak XRD peaks could be observed between the MgO and the Al_2O_3 ones (Fig. S1b). The thermogravimetric analysis (TGA) and derivative thermogravimetry analyses data (Fig. S2) revealed differences between the bulk MMO precursors and acetone-treated ones; namely, the temperatures of the weight loss stages of the treated precursors were lower than those of the bulk precursors because the treated nanosheets exfoliated to be thinner.[49] The N_2 adsorption–desorption isotherms at 77 K and the DFT pore size distributions of the supports determined using the density functional theory are illustrated in Fig. S3, whereas the calculated textural properties are listed in Table S2. Compared to SBA-15, MMOs exhibited much lower surface areas but comparable pore volume, broader pore size distributions, and higher average pore diameters, which provide good opportunity for the diffusion of CO_2 via the high loading of PEI.

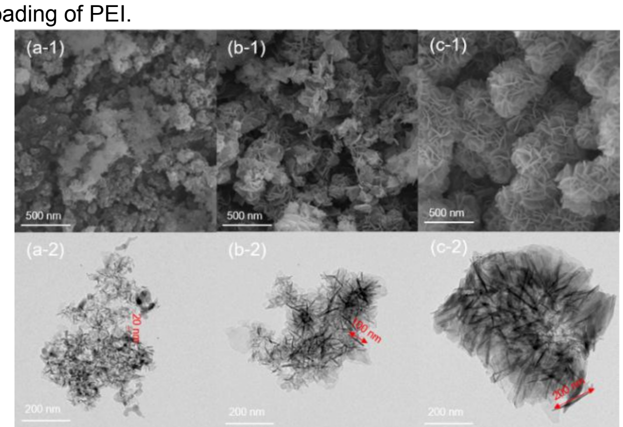

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Fig. 1. Scanning and transmission electron microscopy images of $Mg_xAl\text{-}CO_3$ mixed metal oxide precursors synthesized using the aqueous miscible organic solvent treatment method; (a) x = 0.55, (b) x = 2, and (c) x = 3.

The morphology and element distributions of 0, 33, 50, 67 wt.% PEI-impregnated $Mg_{0.55}$Al-O (denoted as PEIy/$Mg_{0.55}$Al-O, y = 0, 33, 50, 67) were studied (Fig. 2). After vacuum thermal treatment, the exfoliated $Mg_{0.55}$Al-O nanosheets aggregated into spherical particles approximately 100 nm in diameter. At the low PEI loading of 33 wt.%, the impregnated PEI preferentially attached to the sides of the MMO layers, which led to the formation of thicker and larger nanosheets and the gradual increase in particle size. Abundant slit-like channels for CO_2 diffusion remained when the PEI loading reached 50 wt.%. At higher PEI loadings, which exceeded the pore volume of the support, the MMO nanosheets started to be covered by PEI. However, the stacked PEI still presented porous morphology owing to the use of $Mg_{0.55}$Al-O framework. The energy dispersive spectroscopy (EDS) results indicated that the impregnated PEI was distributed uniformly in all the three PEI/$Mg_{0.55}$Al-O composites.

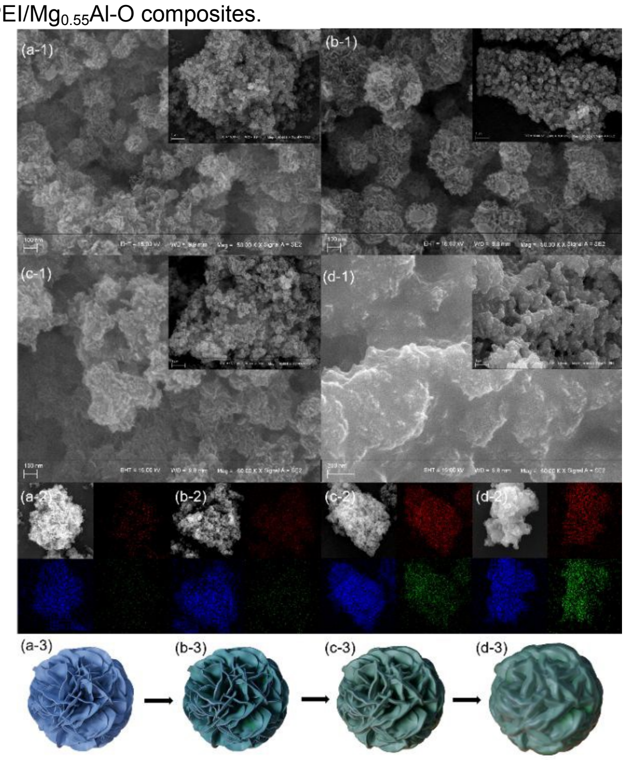

Fig. 2. Scanning electron microscopy images and energy dispersive X-ray spectroscopy (EDS) results of (a) $Mg_{0.55}$Al-O, (b) PEI33/$Mg_{0.55}$Al-O, (c) PEI50/$Mg_{0.55}$Al-O, and (d) PEI67/$Mg_{0.55}$Al-O. The red, blue, and green spots in the EDS scans represent Mg, Al, and N, respectively.

The CO_2 adsorption isotherm of the 0, 33, 50, and 67 wt.% PEI-impregnated $Mg_{0.55}$Al-O and SBA-15 at 25 °C was studied (Fig. 3a). The capacity of PEI33/$Mg_{0.55}$Al-O at 0.4 mbar (1.080 mmol g^{-1}) was comparable to that of PEI33/SBA-15, and was higher than that of PEI33/γ-Al_2O_3 and PEI33/MgO (Fig. S4). Interestingly, the capacities of PEI33/Mg_2Al-O and PEI33/Mg_3Al-O were unsatisfactory, despite their supports presenting surface areas and pore distributions similar to that of PEI33/$Mg_{0.55}$Al-O (Table S2). It was therefore concluded that the morphology of $Mg_{0.55}$Al-O which featured small plate sizes could play a critical role in enhancing the adsorption efficiency of the impregnated PEI. As the PEI loading ratio increased, the CO_2 uptake of PEI/$Mg_{0.55}$Al-O at 0.4 mbar increased linearly whereas the amine efficiency of PEI/SBA-15 started to decrease (Fig. S5). Remarkably, the CO_2 uptake of PEI67/$Mg_{0.55}$Al-O reached 2.272 mmol g^{-1}. This was one of the highest values among class 1 adsorbents in literatures for DAC under dry conditions according to our knowledge. Although larger capacities have been reported for other type of adsorbents (*e.g.* 3 mmol g^{-1} for mmen-appended Mg_2(dobpdc)[53]), the low cost, facile fabrication as well as promising amine efficiency at high PEI loadings without additives rendered PEI67/$Mg_{0.55}$Al-O attractive for scalable direct air capture applications. In addition, the near zero initial capacities of the supports indicated that almost all the CO_2 adsorption sites under ultradilute conditions were attributed to the chemisorption of PEI. The performance of PEI67/$Mg_{0.55}$Al-O at higher adsorption temperatures was investigated (Fig. S6), and the heat of adsorption of PEI67/$Mg_{0.55}$Al-O was evaluated to be 75.3 kJ mol^{-1} using the Clausius−Clapeyron equation (Fig. S7).

The CO_2 adsorption kinetics of the PEI/SBA-15 and PEI/$Mg_{0.55}$Al-O were compared using TGA tests at 25 °C under 400 ppm CO_2 in N_2 atmosphere (Fig. 3b). All samples exhibited fast initial CO_2 adsorption followed by a slow diffusion-controlled stage during the remaining duration of the test (600 min). The fast uptakes of PEI/$Mg_{0.55}$Al-O during the first stage, which was surface reaction rate-limited, were larger than that of PEI/SBA-15 for all the three analyzed PEI loadings. The temperature programmed desorption (TPD) results in Fig. S8 revealed that stronger adsorption sites existed on PEI/$Mg_{0.55}$Al-O than on PEI/SBA-15. The results of the N_2 adsorption–desorption tests, which were conducted at 77 K, provided further insight into the changes in the structure of the supports after PEI impregnation (Fig. 3c, d, and Table S3). Despite its initial surface area being larger than that of SBA-15, $Mg_{0.55}$Al-O effectively prevented pore blockage and presented more reasonable pore size distributions, larger surface area, and pore volume for CO_2 diffusion after PEI loading. Therefore, it was concluded that the use of $Mg_{0.55}$Al-O promoted the uniform distribution of the impregnated PEI on the surface of the support, which facilitated the accessibility of CO_2 molecules to the exposed strong adsorption sites and promoted the CO_2 adsorption kinetics in ultradilute conditions.

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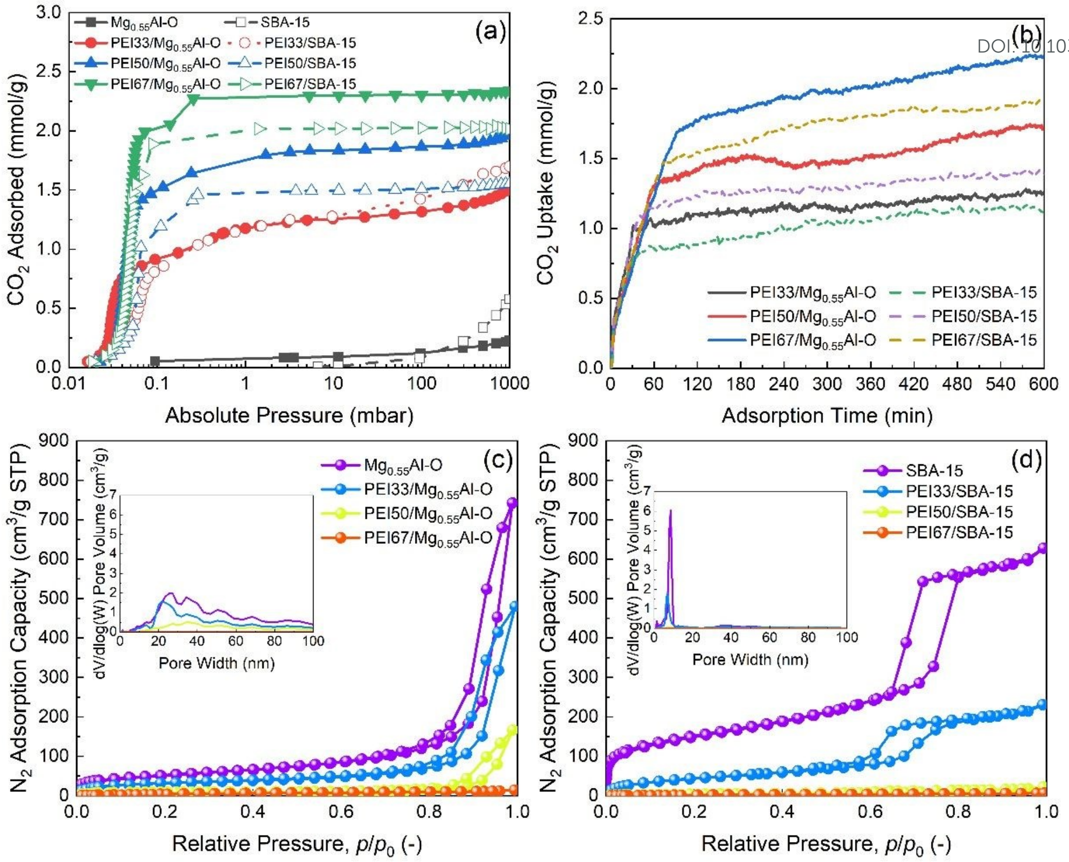

Fig. 3. Characterization of CO_2 adsorption performance of poly(ethylenimine) (PEI)-based adsorbents in ultradilute conditions: (a) CO_2 adsorption isotherm of PEI/SBA-15 and PEI/$Mg_{0.55}$Al-O at 25 °C, (b) CO_2 adsorption kinetics of PEI-impregnated SBA-15 and $Mg_{0.55}$Al-O at 25 °C under 400 ppm CO_2 in N_2. N_2 adsorption-desorption isotherms at 77 K and pore distributions of PEI-impregnated (c) $Mg_{0.55}$Al-O and (d) SBA-15 with different PEI loadings calculated by density functional theory.

Amine-functionalized CO_2 adsorbents, particularly class 1 adsorbents without aminosilanes linking, commonly face stability problems.[43] However, the thermal decomposition results depicted in Fig. 4a and Fig. S9 provided strong evidence for the interactions between the impregnated amines and supports. Two weight loss peaks, which were 350 °C apart, were present in the curve of PEI/SBA-15. The low-temperature peak was related to the decomposition of the PEI dispersed in the pores of SBA-15 and the high-temperature peak was ascribed to the breaking of the hydrogen bonds between the remaining amine and silanols on the support.[44] Both peaks shifted toward higher temperatures as the PEI loadings increased, and this was attributed to the agglomeration of PEI into larger particles. In contrast, only the peak above 350 °C was observed in the curves of all PEI/$Mg_{0.55}$Al-O samples and the peak temperature did not change significantly, which suggested the presence of strong interactions between PEI and $Mg_{0.55}$Al-O. It was demonstrated that the CO_2 uptake of PEI67/SBA-15 rapidly decreased after it underwent desorption above 200 °C, whereas the uptake of PEI67/$Mg_{0.55}$Al-O did not decrease until the temperature reached 300 °C (Fig. 4b). Moreover, a slight increase in the CO_2 uptake of PEI67/$Mg_{0.55}$Al-O was observed at higher desorption temperatures, which could be ascribed to the enhanced kinetics owing to the presence of the optimized CO_2 diffusion channel and increased surface area after moving the weakly attached PEIs (Fig. S10). It has been demonstrated that coordinatively unsaturated oxygens or metals were created via the insertion of Al^{3+} ions into MMOs with periclase MgO lattice or the escape of Al atoms from layers during calcination, and their density increased as the lateral size of the MMO layers decreased.[54,55] Therefore, the defect-abundant $Mg_{0.55}$Al-O nanosheets enabled to facilitate the strong electrostatic attractions between the impregnated PEI and MMO layers, which rendered the PEI/$Mg_{0.55}$Al-O composites thermally robust (Fig. 4c).

To further verify the stability of PEI67/$Mg_{0.55}$Al-O, cyclic breakthrough tests were conducted using a home-made fixed bed microreactor system (Fig. S11). The capacity of PEI67/$Mg_{0.55}$Al-O during at least 20 cycles that consisted of 2 h adsorption at 25 °C under 400 ppm CO_2 in the absence of moisture followed by 1 h desorption at 120 °C under pure N_2 atmosphere was determined to be stable; the slight decrease in performance of 0.007 mmol g^{-1} per cycle was ascribed to the composite coming into contact with highly concentrated CO_2 during the regeneration step via the formation of urea linkages.[56] The CO_2 uptake of PEI67/$Mg_{0.55}$Al-O increased by 50% for wet CO_2 when the inlet gas was passed through a scrubbing bottle (Fig. S12). The supported amines formed ammonium carbamate when exposed to a maximum of 0.5 mol of CO_2/mol of N under dry conditions after the adsorption of CO_2, which was confirmed using the *in situ* Fourier-transform infrared spectroscopy (FTIR) spectra in Fig. 4d. However, it was reported that the CO_2 adsorption mechanism was changed to the formation of ammonium carbonate/bicarbonate when the supported amines were exposed to a maximum of 0.5 mol of CO_2/mol of N in the presence of moisture.[20] The results depicted in Fig. 4e also suggested that $Mg_{0.55}$Al-O efficiently prevented blocking the access

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of amine via water oversaturation at high PEI loadings, which has been commonly reported for silica-based materials.[33] To produce high-purity CO_2 in the practical conditions, regeneration with steam stripping followed by condensing leads to faster desorption kinetics and higher adsorbent lifetime compared to CO_2 purge or vacuum swing.[41] The results in Fig. 4e indicate that there have been no noticeable changes in the performance of the proposed PEI/MMO composites when wet N_2 was used during thermal regeneration, which indicated that steam stripping could be a possible strategy for concentrating the desorbed CO_2. Overall, the PEI/MMO composites exhibited unexpectedly large CO_2 uptake, fast kinetics, and high stability under ultradilute conditions compared to PEI/mesoporous silica materials in literatures (Table 1), thus being highly attractive for DAC.

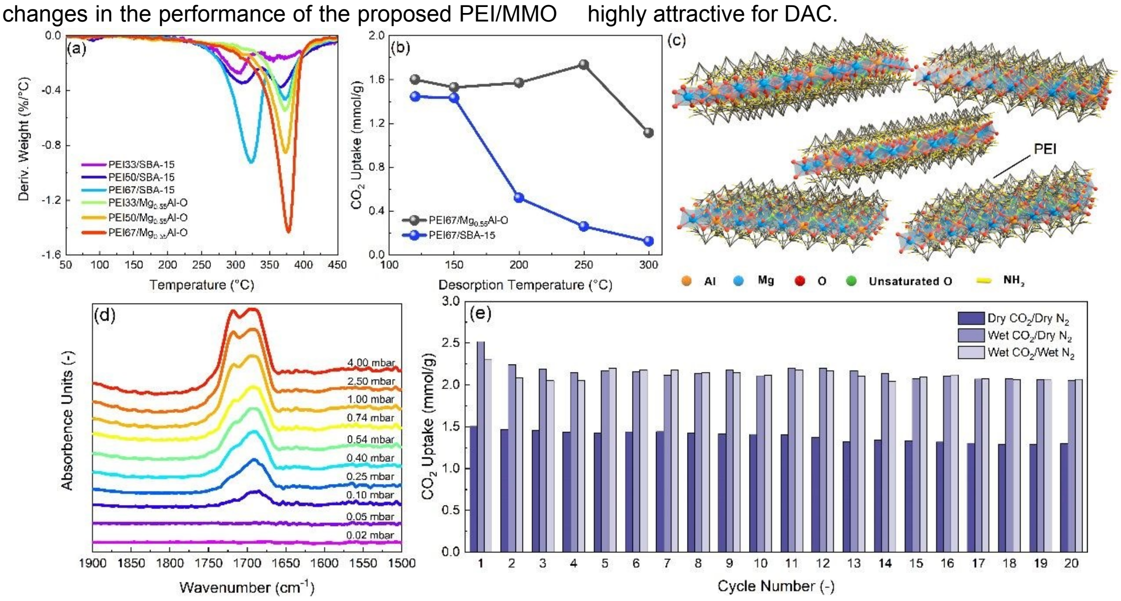

Fig. 4. Stability of poly(ethylenimine) (PEI)-impregnated samples under direct air capture conditions: (a) thermal stability from 50 to 450 °C at a heating rate of 5 °C min^{-1} under N_2 flow, (b) CO_2 uptake in 2 h at 400 ppm CO_2 and 25 °C after desorption at different temperatures, (c) schematic diagram of PEI/MMOs structure (MMOs denotes layered double hydroxide-derived mixed metal oxides), (d) *in situ* infrared spectroscopy for the CO_2 adsorption of PEI67/$Mg_{0.55}$Al-O at 25 °C, and (e) cyclic stability of PEI67/$Mg_{0.55}$Al-O with 2 h adsorption at 25 °C followed by 1 h desorption at 120 °C.

Table 1. PEI-impregnated supports for direct air capture; Uptakes, kinetics, and stabilities are presented.

Support	PEI (M_w)	Loadings (wt.%)	Adsorption conditions	Desorption conditions	Method	Capacity (mmol g^{-1}[c])	Kinetics[d] (mmol g^{-1} h^{-1})	Cycle	Loss (%)	Ref
Commercial silica	800	45	400 ppm CO_2/Ar at 25 °C	Pure Ar at 110 °C	TGA	2.36	0.24	4	30	32
Fumed silica	25,000	50	400 ppm CO_2/21% O_2/79% N_2 at 25 °C	21 O_2/79% N_2 at 85 °C	TGA/PB	1.40	0.70	4	0	31
Fumed silica	25,000	50	400 ppm CO_2/21% O_2/79% N_2 at 35 °C	21 O_2/79% N_2 at 85 °C	TGA	1.52	0.84			31
Fumed silica	25,000	50	400 ppm CO_2/21% O_2/79% N_2 at 45 °C	21 O_2/79% N_2 at 85 °C	TGA	1.12	0.66			31
Fumed silica	25,000	50	400 ppm CO_2/21% O_2/79% N_2 at 25 °C	Under vacuum at 85 °C	PB[b]	1.71 1.41 (wet)		4 4	0 0	33
Zr-SBA-15	800	35	400 ppm CO_2/Ar at 25 °C	Pure Ar at 110 °C	TGA	0.85	0.50	4	2.2	34,35
SBA-15	800	30	400 ppm CO_2/He at 30 °C	Pure He at 110 °C	TGA	0.94				36
SBA-15	800	50	400 ppm CO_2/He at 30 °C	Pure He at 110 °C	TGA	1.58	0.89			37,39
SBA-15+ PEG200[a]	800	50	400 ppm CO_2/He at 30 °C	Pure He at 110 °C	TGA	1.64	0.96			38
Silica fiber	800	26	380 ppm CO_2/N_2 at 35 °C	Pure N_2 at 90 °C	PB	0.62 1.60 (wet)		20 6	9.7 7	23
PE-MCM-41	800	40	400 ppm CO_2/N_2 at 25 °C	Pure N_2 at 90 °C	PB	2.18 2.92 (wet)		20 20	4.6 2.1	40

MCF	1,200	63	420 ppm CO_2/N_2 at 46 °C	Pure N_2 at 100 °C	PB	1.94 2.52 (wet)				41
SBA-15	600	50	400 ppm CO_2/N_2 at 25 °C	Pure N_2 at 120 °C	TGA/PB	1.46	0.80			This work
SBA-15	600	67	400 ppm CO_2/N_2 at 25 °C	Pure N_2 at 120 °C	TGA/PB	1.92	1.0			This work
MMO	600	50	400 ppm CO_2/N_2 at 25 °C	Pure N_2 at 120 °C	TGA/PB	1.66	0.91			This work
MMO	600	67	400 ppm CO_2/N_2 at 25 °C	Pure N_2 at 120 °C	TGA/PB	2.27 - (wet)	1.1	20 20	14 2.7	This work

[a] PEG200: MW 200 poly(ethylene glycol). [b] PB: Packed bed. [c] Based on sorbent material. [d] CO_2 uptake in the first 90 min.

Conclusions

We present a new family of PEI-functionalized Mg-Al-CO_3 MMO class 1 adsorbents for CO_2 capture under ultradilute conditions. The low Mg/Al ratios of Mg_xAl-CO_3 and the use of the AMOST method lead to the exfoliation of the MMO precursors into nanosheets with plate sizes in the range of 20–200 nm, which self-assembled into spherical particles upon thermal treatment. The impregnated polyamines dispersed uniformly on the surface of the MMO nanosheets, and thus, they created optimal accessibility for the CO_2 molecules to the exposed strong adsorption sites under ultradilute conditions. At high PEI loading ratios of up to 67 wt.%, the polyamines started to cover the MMO frameworks but the porous structure was still maintained and their amine efficiency was not significantly decreased. The proposed PEI/MMO composites exhibited higher CO_2 uptake and better kinetics than PEI/SBA-15 owing to the ideal morphologies and nanostructures of their supports. Remarkably, the 67 wt.% PEI-impregnated $Mg_{0.55}$Al-O MMO reaches the total CO_2 uptake of 2.27 mmol g^{-1} and the fast adsorption rate of 1.1 mmol g^{-1} h^{-1} within 90 min at 25 °C under 0.4 mbar CO_2. In addition, the defect-abundant MMO layers provide strong electrostatic attraction to the attached polyamines, which contributed to the robustness of the hybrid adsorbents for at least 20 cycles in the absence and presence of moisture. Overall, the low cost, facile fabrication, and promising CO_2 capacities, kinetics, and thermal and hydrothermal stabilities rendered amine-functionalized MMOs attractive for scalable CO_2 capture applications under ultradilute conditions.

Conflicts of interest

There are no conflicts to declare.

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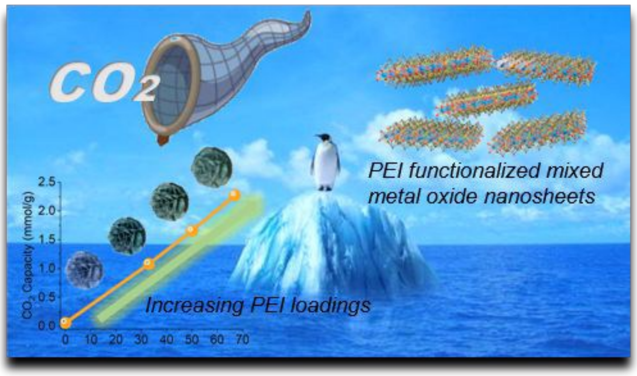

A new family of class 1 adsorbents for direct air capture of CO_2 is prepared using mixed metal oxide nanosheets functionalized with branched poly(ethylenimine). The hybrid adsorbents provide slit-like morphologies and defect-abundant nanostructures to achieve promising adsorption capacities, kinetics, thermal stability, and hydrothermal stability under ultredilute conditions.